KUBERNETES

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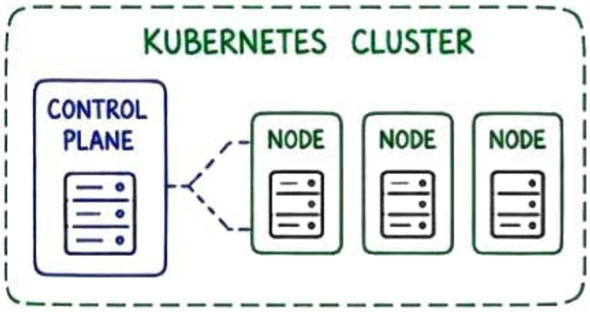
1. WHAT IS KUBERNETES?

Kubernetes is an open-source container orchestration platform. It automates deployment, scaling and management of containerised applications.

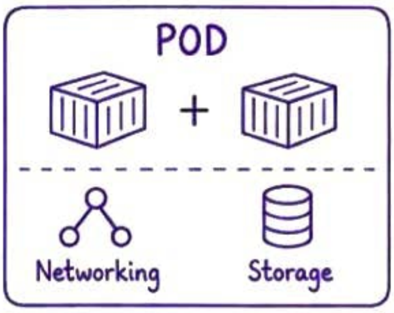
2. WHAT IS A KUBERNETES CLUSTER?

A cluster is a group of machines running Kubernetes. It contains a control plane and worker nodes.



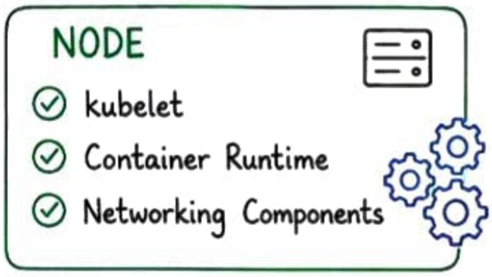
3. WHAT IS A POD?

A Pod is the smallest deployable unit in Kubernetes. It contains one or more containers that share networking and storage.



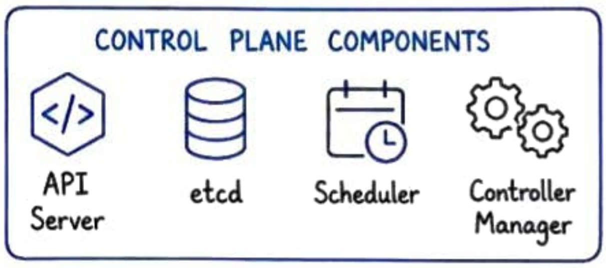
4. WHAT IS A NODE?

A Node is a physical or virtual machine that runs Pods. It includes kubelet, a container runtime and networking components.



5. WHAT DOES THE CONTROL PLANE DO?

The control plane maintains the desired cluster state. Main components: API server, etcd, scheduler and controller manager.



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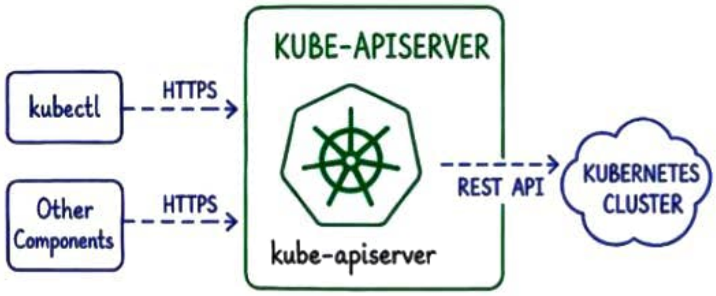
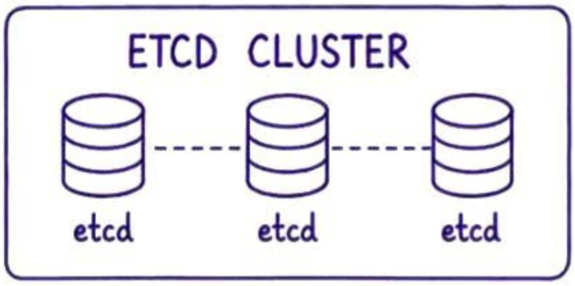
6. WHAT IS KUBE-APISERVER?

The kube-apiserver is the front door of the Kubernetes control plane, kubectl and other components communicate with the cluster through its REST API.



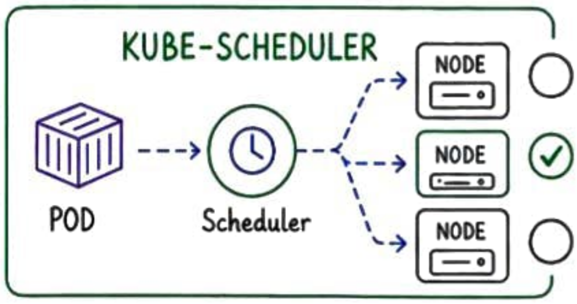
7. WHAT IS ETCD?

etcd is a distributed key-value store. Kubernetes uses it to store cluster configuration, state and metadata. Regular backups are essential.



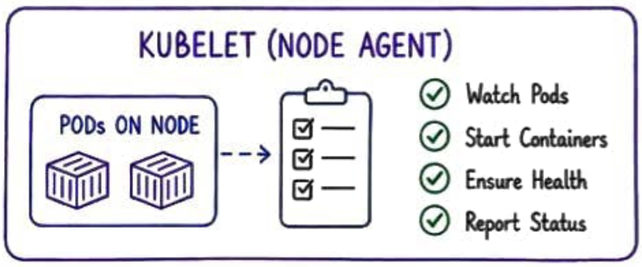
8. WHAT DOES KUBE-SCHEDULER DO?

The scheduler selects a suitable node for each unscheduled Pod. It checks resources, constraints, affinity rules and taints.



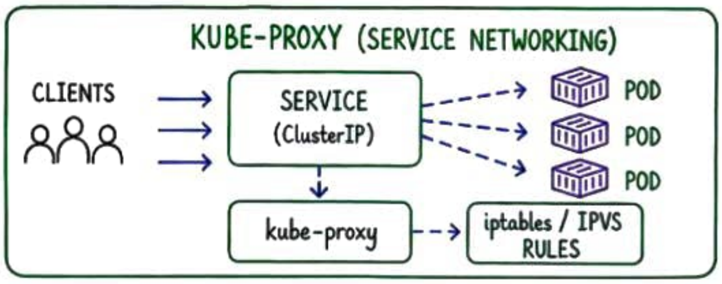
9. WHAT IS KUBELET?

kubelet is the node agent. It watches Pods assigned to its node and ensures the required containers are running and healthy.



10. WHAT IS KUBE-PROXY?

kube-proxy implements Service networking on nodes, commonly using iptables or IPVS rules.



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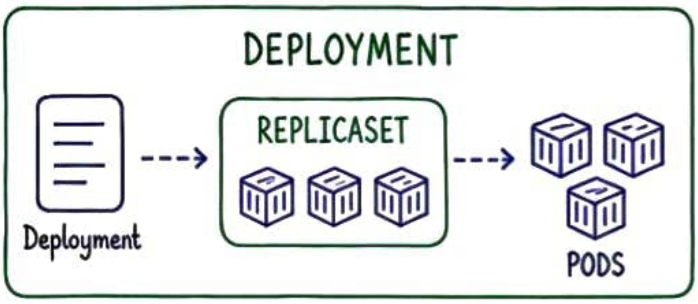
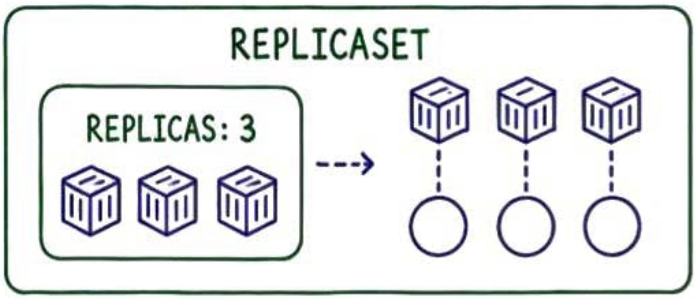
11. WHAT IS A DEPLOYMENT?

A Deployment manages stateless application Pods through ReplicaSets. It supports scaling, rolling updates, rollout history and rollback.



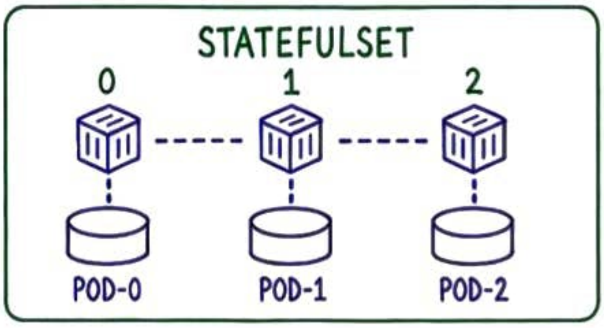
12. WHAT IS A REPLICASET?

A ReplicaSet ensures that a specified number of identical Pod replicas are running. It is usually managed by a Deployment.



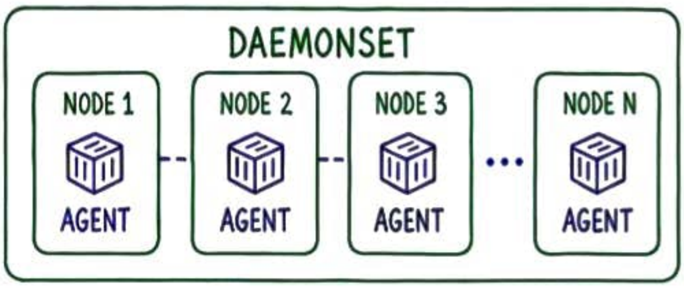
13. WHAT IS A STATEFULSET?

A StatefulSet manages stateful Pods that need stable identities, ordered deployment and persistent storage. It is suitable for databases.



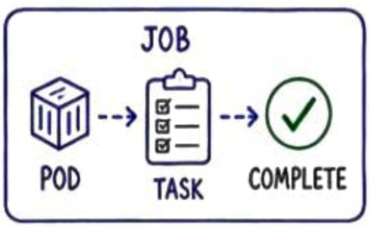
14. WHAT IS A DAEMONSET?

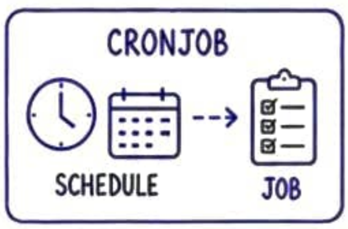
A DaemonSet runs one copy of a Pod on every eligible node. It is commonly used for log collectors, monitoring agents and networking components.



15. WHAT ARE JOB AND CRONJOB?

A Job runs Pods until a task completes successfully. A CronJob creates Jobs on a schedule, such as backups or periodic reports.





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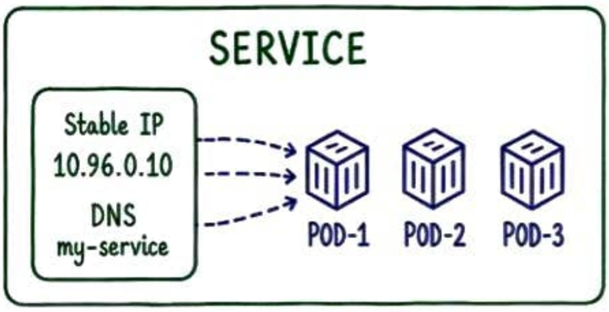
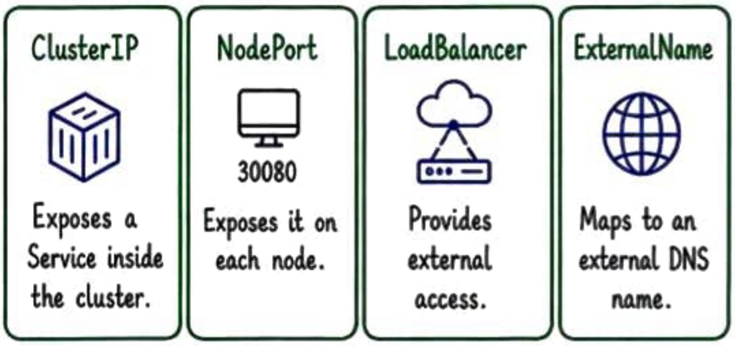
16. WHAT IS A KUBERNETES SERVICE?

A Service gives a stable virtual IP and DNS name to a changing set of Pods. It selects backend Pods using labels.



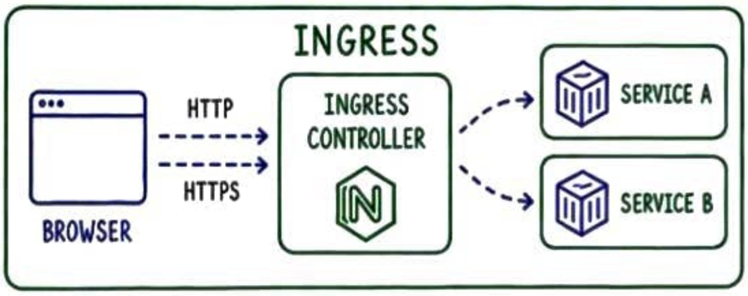
17. EXPLAIN SERVICE TYPES.

ClusterIP exposes a Service inside the cluster. NodePort exposes it on each node. LoadBalancer provides external access. ExternalName maps to an external DNS name.



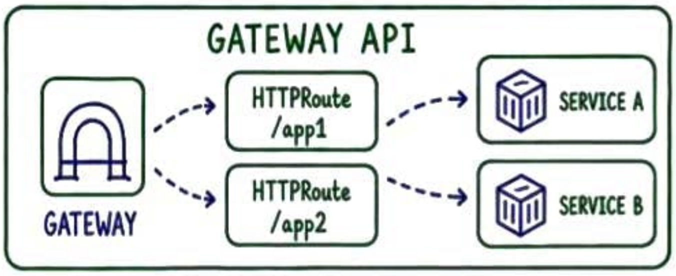
18. WHAT IS INGRESS?

Ingress defines HTTP and HTTPS routing from outside the cluster to Services. An Ingress controller implements the routing rules.



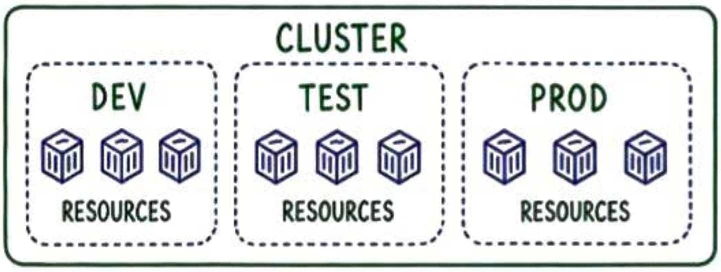
19. WHAT IS THE GATEWAY API?

Gateway API is a newer, role-oriented model for traffic routing. It provides more expressive and extensible resources than traditional Ingress.



20. WHAT IS A NAMESPACE?

A Namespace logically separates resources within a cluster. It helps organise teams and environments using RBAC, quotas and network policies.



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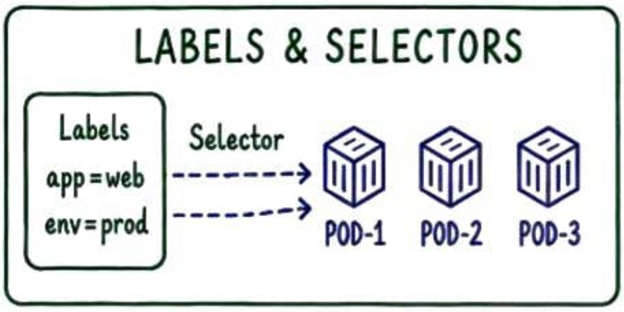
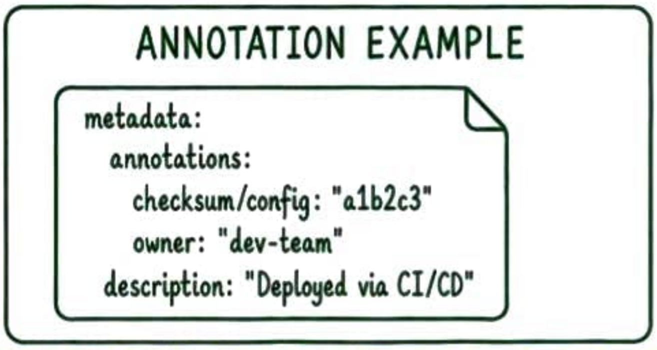
21. WHAT ARE LABELS AND SELECTORS?

Labels are key-value metadata attached to objects. Selectors use labels to group or target resources, such as when a Service finds its Pods.



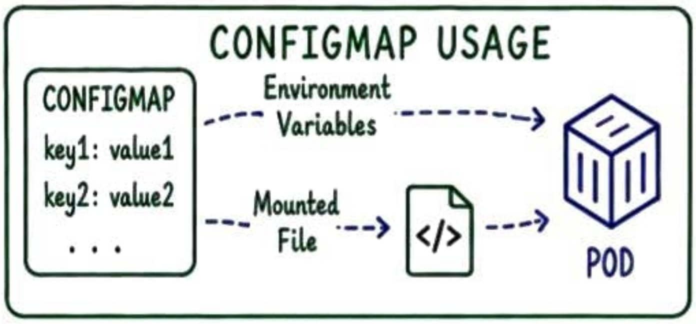
22. WHAT ARE ANNOTATIONS?

Annotations store non-identifying metadata, such as tool information, checksums or external references. They are not used to select objects.



23. WHAT IS A CONFIGMAP?

A ConfigMap stores non-sensitive configuration as key-value data or files. Pods can consume it as environment variables or mounted volumes.

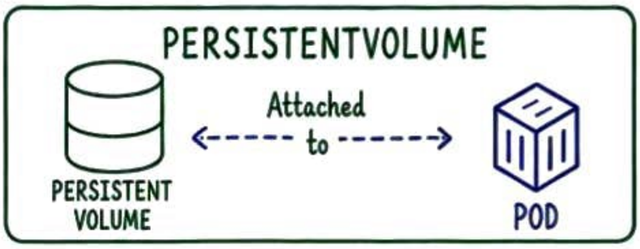


24. WHAT IS A SECRET?

A Secret stores sensitive values such as tokens or passwords. Base64 encoding is not encryption, so apply encryption and strict RBAC.

25. WHAT IS A PERSISTENTVOLUME?

A PersistentVolume is cluster storage provisioned by an administrator or dynamically by a StorageClass. Its lifecycle is independent of a Pod.



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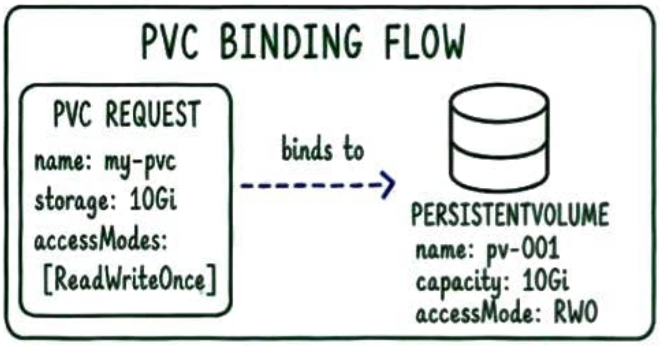
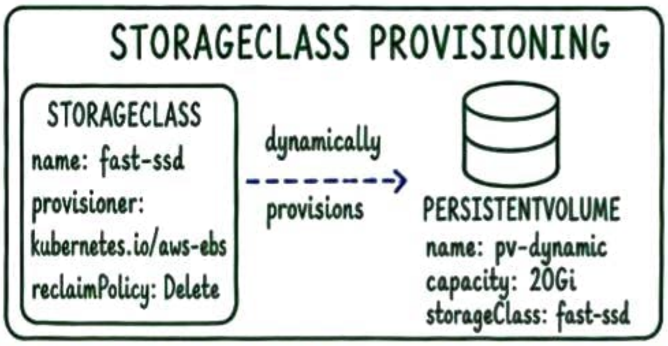
26. WHAT IS A PERSISTENTVOLUMECLAIM?

A PersistentVolumeClaim is a user's request for storage with a particular size and access mode. Kubernetes binds it to a suitable PersistentVolume.



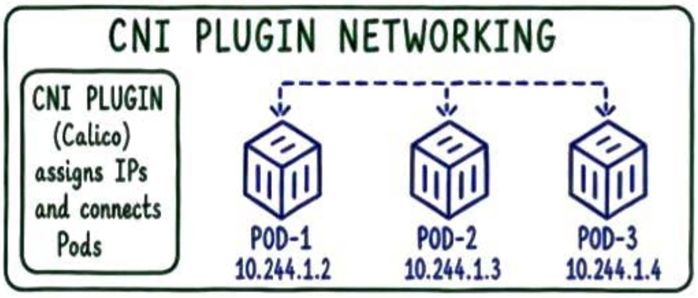
27. WHAT IS A STORAGECLASS?

A StorageClass describes a class of storage and its provisioner. It enables dynamic creation of PersistentVolumes when claims are made.



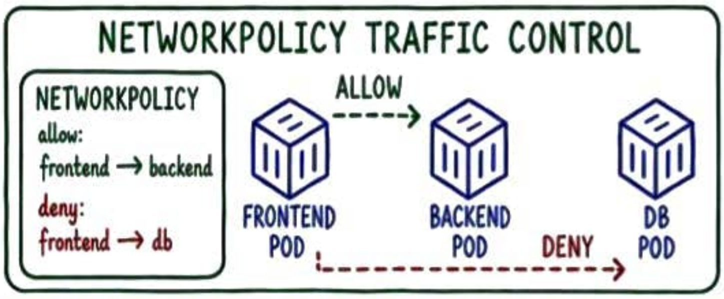
28. WHAT IS A CNI PLUGIN?

A Container Network Interface plugin provides Pod networking and IP allocation. Examples include Calico, Cilium and Flannel.



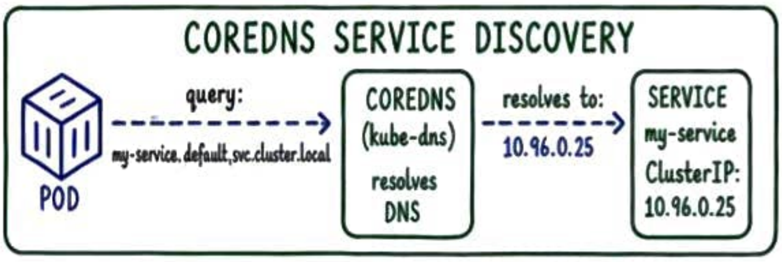
29. WHAT IS A NETWORKPOLICY?

A NetworkPolicy controls allowed traffic to and from selected Pods. It works only when the installed network plugin supports policy enforcement.



30. WHAT IS COREDNS?

CoreDNS provides DNS-based service discovery in a Kubernetes cluster. Pods use it to resolve Service names and external domains.



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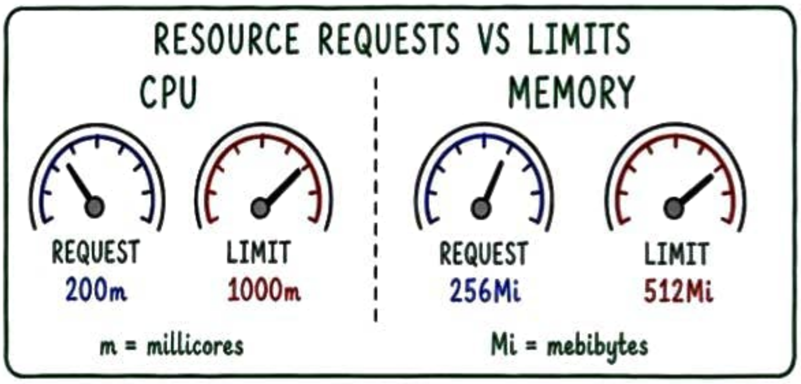
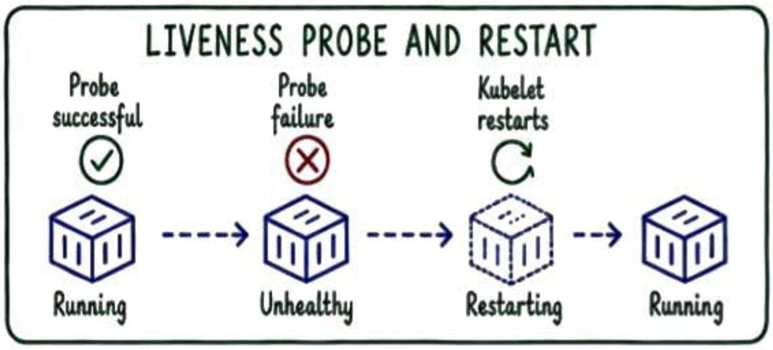
31. WHAT ARE RESOURCE REQUESTS AND LIMITS?

Requests reserve expected CPU or memory and influence scheduling. Limits cap resource use. Excess memory can cause an OOM kill, while CPU is usually throttled.



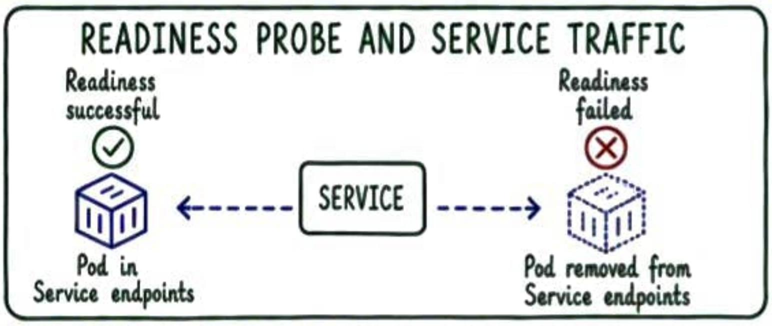
32. WHAT IS A LIVENESS PROBE?

A liveness probe checks whether a container is functioning. Repeated failure causes kubelet to restart the container.



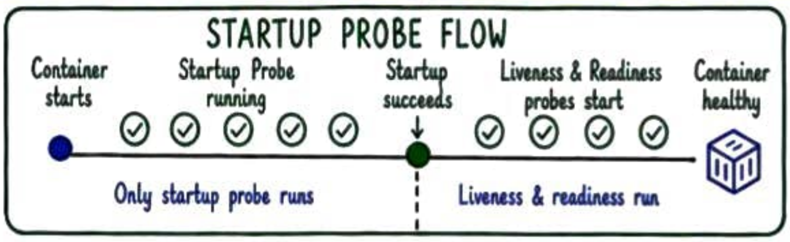
33. WHAT IS A READINESS PROBE?

A readiness probe checks whether a Pod can accept traffic. If it fails, the Pod is removed from Service endpoints without being restarted.



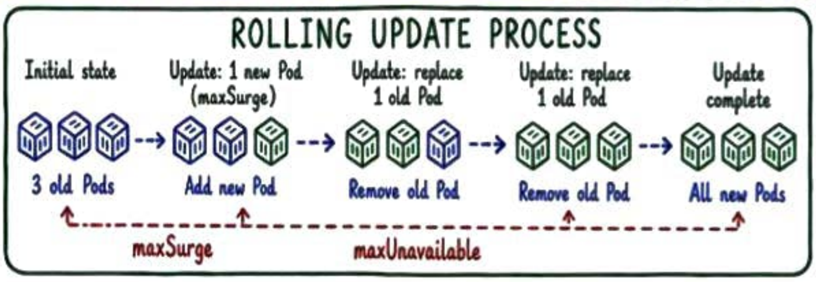
34. WHAT IS A STARTUP PROBE?

A startup probe protects slow-starting containers. Liveness and readiness checks begin only after the startup probe succeeds.



35. WHAT IS A ROLLING UPDATE?

A rolling update gradually replaces old Pods with new Pods while maintaining availability. maxSurge and maxUnavailable control the rollout.



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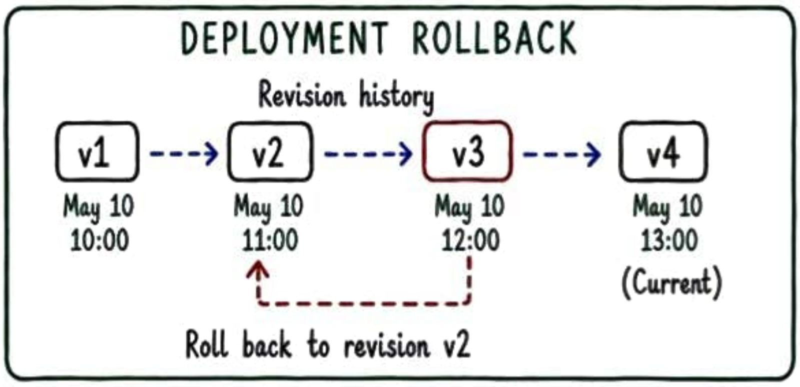
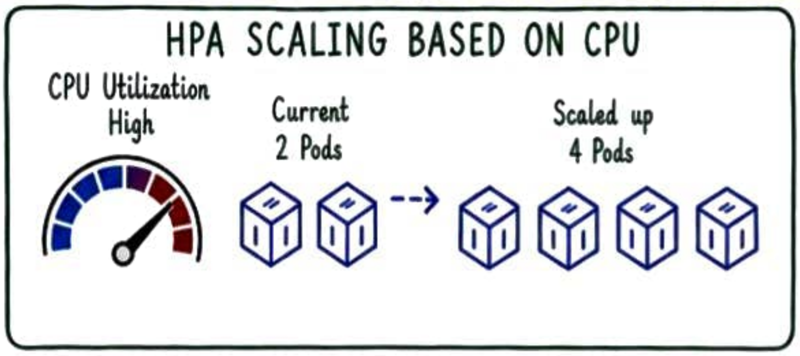
36. HOW DO YOU ROLL BACK A DEPLOYMENT?

Check history with `kubectl rollout history deployment/name`. Roll back using `kubectl rollout undo deployment/name`. Specify a revision when required.



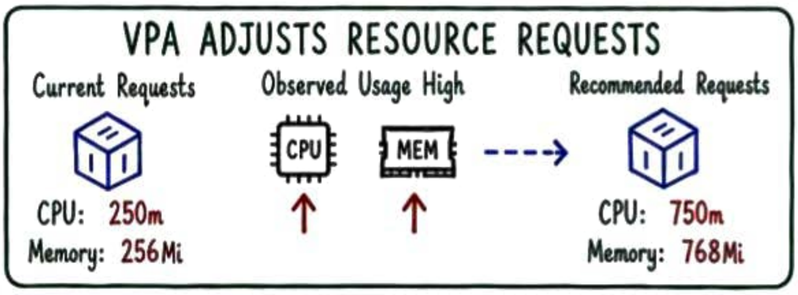
37. WHAT IS HORIZONTAL POD AUTOSCALER?

HPA automatically changes the number of Pod replicas using CPU, memory or custom metrics. Accurate resource requests and metrics are important.



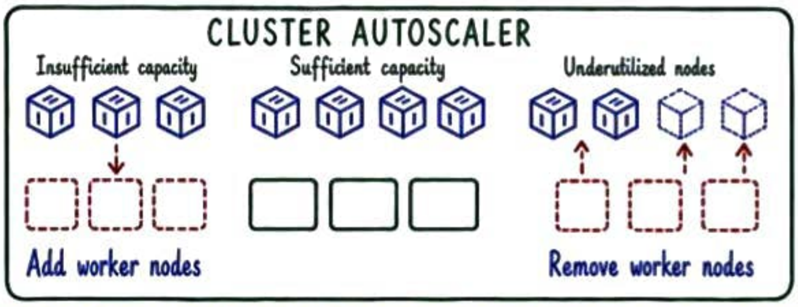
38. WHAT IS VERTICAL POD AUTOSCALER?

VPA recommends or adjusts Pod CPU and memory requests based on usage. Applying changes may require Pod recreation.



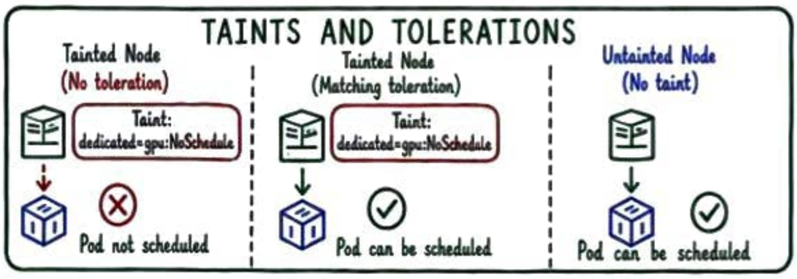
39. WHAT IS CLUSTER AUTOSCALER?

Cluster Autoscaler adds nodes when Pods cannot be scheduled due to capacity and removes underused nodes when workloads can be moved safely.



40. WHAT ARE TAINTS AND TOLERATIONS?

A taint repels Pods from a node. A matching toleration allows a Pod to be scheduled there, but does not guarantee placement.



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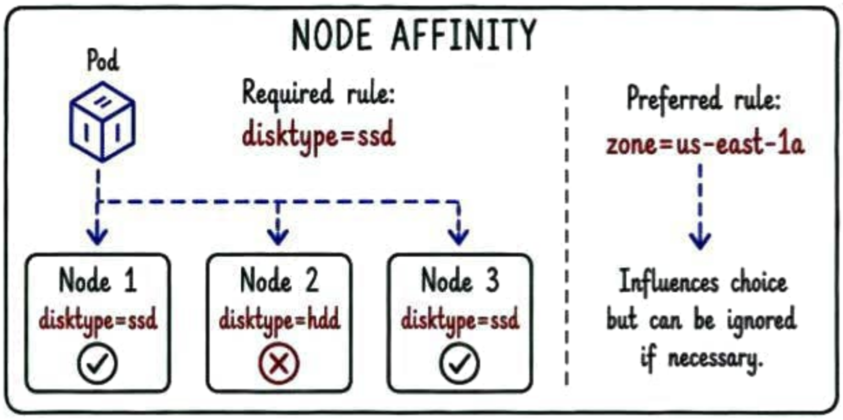
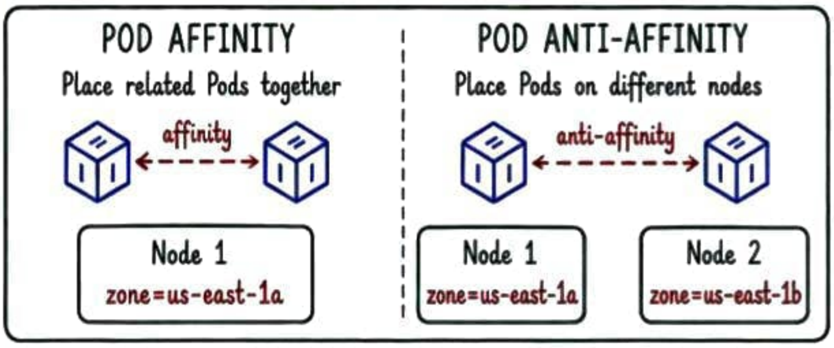
41. WHAT IS NODE AFFINITY?

Node affinity places Pods based on node labels. Required rules are mandatory. Preferred rules influence scheduling but can be ignored if necessary.



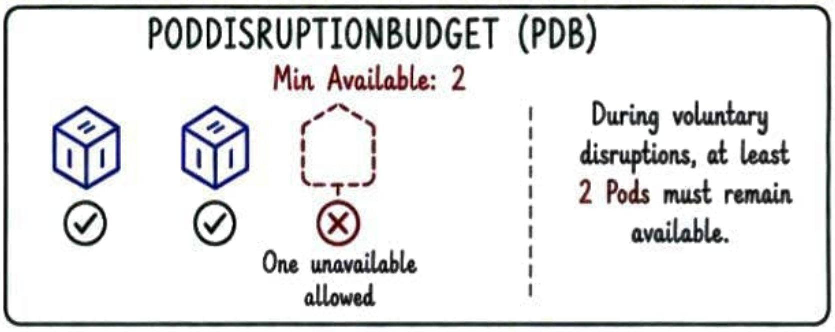
42. WHAT ARE POD AFFINITY AND ANTI-AFFINITY?

Pod affinity places Pods near matching Pods. Pod anti-affinity separates them for better availability across nodes or zones.



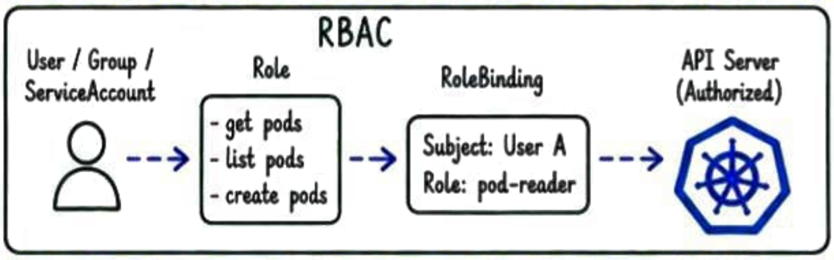
43. WHAT IS A PODDISRUPTIONBUDGET?

A PodDisruptionBudget limits unavailable replicas during voluntary disruptions such as node drains. It does not prevent failures or all evictions.



44. WHAT IS RBAC?

Role-Based Access Control authorises API actions. Roles define permissions. RoleBindings assign them to users, groups or ServiceAccounts.



45. WHAT IS A SERVICEACCOUNT?

A ServiceAccount gives a workload an identity inside Kubernetes. Pods use its short-lived token to access the API according to RBAC permissions.